



Leveraging LLMs for Written to Spoken Style Data Transformation to Enhance Spoken Dialog State Tracking

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Abstract

Dialog State Tracking (DST) is an important part of Task-Oriented Dialog (TOD) systems, as it needs to navigate the complex human conversational flow to accomplish a task. Most TOD systems are trained on written-style text data, and their performance plunges when deployed in spoken scenarios due to natural disfluencies and human-speech recognition errors. Labeled spoken-style TOD data is limited because of the high data collection cost and privacy concerns. As Large Language Models (LLMs) emerge as a tool for synthetic text data generation, we explored their capability to generate spoken-style text-based TOD data. Through meticulously crafting LLM prompts, our generated labeled spoken style TOD data improved the absolute Joint Goal Accuracy (JGA) by 3.39% and relative JGA by 11.6%, for dedicated DST models. In this work, we showcase our divide-and-conquer-based data generation strategies and DST training to improve the performance of task-specific dialog models.

Index Terms: Spoken Dialog Systems, Task-Oriented Dialog Systems, Data Generation, Spoken Dialog State Tracking

1. Introduction

The advancements in Natural Language Processing (NLP) have led dialog models to achieve impressive performance on various tasks like reasoning-based question answering [1], recommendation systems [2], and performing tasks on the user's behalf through Task-Oriented Dialog (TOD) systems [3]. Dialog State Tracking (DST) is a central task in TOD systems that detects the user's intention, collects the relevant information, and keeps track of the dialog history in form of a dialog state, in order to take the next appropriate action [4]. Generally, TOD models are pre-trained on a large corpus of unlabeled or pseudo-labeled text data and then fine-tuned for a specific task using labeled text data [4, 5]. However, deploying such models in spoken scenarios poses unique challenges like differences in spoken and written style, and task mishandling because of speech recognition errors under environmental noise [6]. Moreover, speech disfluencies and uninventable complex flow of the spoken conversation comprising of multiple turns and incomplete information in a single turn requires additional consideration to infer the correct and complete information from the user's utterance [7].

Adapting text-based TOD models to spoken scenarios requires dialog speech and fine-grained task labels, which is an expensive approach and raises privacy concerns linked with human speech data. Some works have tried to tackle this problem with synthesized speech-paired data to adapt the models to spoken scenarios [8, 9]. However, synthesized speech lacks natural disfluencies and indirect reasoning occurring in human speech [10], and it is also a relatively expensive approach re-

quiring significant computation to generate dense speech signals. This motivated us to explore the feasibility of text-based spoken style data generation as it is a significantly more economical approach than speech synthesis or human-labeled data collection.

As the generalization capabilities of Large Language Models (LLMs) improve, they are also utilized for text data synthesis using few-shot prompts [11]. Although a synthetic data generation approach has been utilized for TOD data generation [12, 13], the capability of LLMs is yet to be tested for spoken-style data generation. In this work, we exploited the generative capability of LLMs to generate spoken-style speech transcripts in text format to use as additional training data for adapting task-based text dialog models in spoken scenarios. There are multifold benefits to utilizing LLMs for generating dialog data. Firstly, recent LLMs have a strong reasoning ability to generate large-scale data by using just a few reference examples. Furthermore, as the context size of LLMs has significantly increased, it is possible to generate long dialogs with complex dialog flows. Lastly, synthesizing speech data is at least 8X more expensive than text-based speech transcript generation [14]. This motivated us to explore the viability of spoken-style text data generation to bridge the gap between spoken and written dialog systems.

When beginning to generate spoken-style TOD data, utilizing the existing written-style TOD data is an economical approach as TOD fine-grained labels can be utilized as it is, while transforming the style of corresponding dialog utterances. In our experiments, we broke down the spoken-style TOD data generation pipeline into two parts; (1) the part of the dialog requiring a change in style from written to spoken and utilizing the existing TOD labels, and (2) the part that requires the generation of new data having multi-turn dialog along with TOD labels, to collect information from the user. We combine the generated data to form a final multi-domain and multi-turn spoken TOD data to use for the training of a small language model-based TOD system. We designed various prompts and demonstrated how LLM can be leveraged to gradually introduce spoken attributes in written-style data using a step-by-step approach, to generate spoken-style text TOD data.

Our LLM synthesized spoken-style TOD data led the TOD model to achieve significant enhancement in the dialog understanding task, with 3.39% absolute and 11.6% relative improvement in Joint Goal Accuracy (JGA). Our work proves that LLM-generated speech transcripts are viable alternatives to expensive labeled data collection or TTS-based approaches.

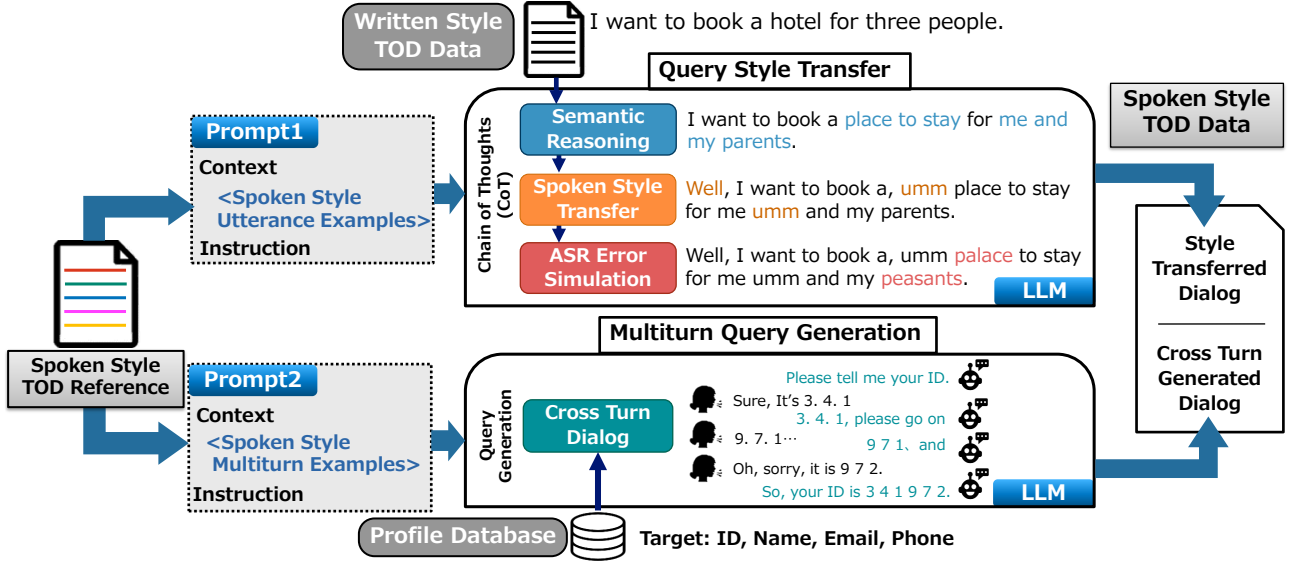


Figure 1: The overall flow of the data generation pipeline. Data generation is divided into two steps, (1) style transfer from written dialogs while keeping the TOD labels intact and (2) generating the new segment of dialog with multi-turn profile information collection from the user.

2. Task-Oriented Dialog (TOD) Systems

TOD systems are specialized dialog models designed to assist users in completing specific tasks while collecting task-specific information through various turns of dialog [3]. Among various approaches to designing these systems, two mainstream approaches are the end-to-end TOD approach [15] and modular TOD approach [16]. The modular architecture of TOD systems includes several key components of which the first one is Dialog State Tracking (DST) as a Spoken Language Understanding (SLU) module, which interprets the user's input, extracts relevant information, and maintains the context in the form of dialog state [17]. The DST module is followed by a policy module and a natural language generation (NLG) module to generate the system's responses. Recently with the advancements in LLMs, prompt-based end-to-end TOD methods are proposed [18], however, we believe that fine-grained dialog understanding is essential for scalable and reliable systems. DST module acts as a steering module for the dialog parsing [19], while for reasoning, next action policy, and response generation, a specialized LLM can be combined with the DST module [19]. Such control is even more important in spoken scenarios where an LLM-based end-to-end approach can steer the dialog in an unintended direction.

2.1. Dialog State Tracking (DST)

In TOD systems, DST plays a vital role in the user's dialog parsing. DST mainly infers three components from the user's query i.e. *user-intent*, *slot-type*, and *slot-value*. Conventional DST methods rely on extractive approaches, meaning they only extract the key values that are present in the user's utterance. Such models fail on queries like, "I want to book a place to stay for me and my parents.". Here the extractive approach is unable to predict the user intent as a *book-hotel*. Therefore, recently generative-based DST approaches have been employed

that leverage the reasoning capability of small language model-based backbones [20].

DST becomes an even more challenging task in spoken scenarios, as the spoken dialog utterance has many disfluencies, indirect information requiring reasoning, and speech misrecognition errors. The following example shows DST labels for an indirect reasoning case in a spoken style query.

Query: "I want to book a place to stay for me and my parents."
user-intent: *book-hotel*, (**slot-type, slot-value:**) (*people*, 3)

Moreover, providing slot information in multiple utterances requires proper tracking as illustrated in the bottom of Figure 1. DST module needs to be adapted to these differences from written to spoken dialog systems. 2 shows an example of a dialog between a user and a system to accomplish hotel booking tasks. The TOD system infers the intent of the user, corresponding slots, and their slot values. The correct intent, slot, and value prediction throughout the flow of the dialog leading to the completion of the task implies that the system has successfully finished its task.

3. Strategies for Spoken TOD Adaptation

Two primary modules in TOD systems need consideration for spoken scenarios: the speech recognition module and the DST module. The former is responsible for converting human speech into a text transcript, while the latter maintains the conversational context and guides the conversation in the targeted direction to accomplish a specific task. Several studies have focused on improving each of these components to enhance the overall performance of TOD systems in spoken scenarios.

Text to Speech (TTS) Synthesis: As the labeled TOD data with corresponding speech is limited, the performance of TOD models reduces significantly when deployed in spoken scenarios. Researchers have made significant efforts to make TOD

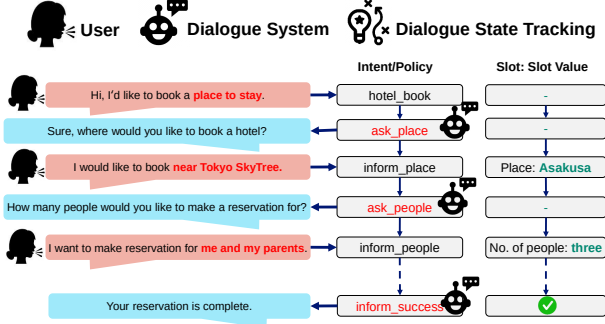


Figure 2: Dialog state tracking module in task-oriented dialog systems, predicting the state of dialog to complete a task.

models robust enough to handle the variability and complexity of human speech by generating end-to-end dialog data by using TTS approach [21]. However, the synthesized speech data is not a reliable surrogate for human speech, as it falls short of replicating the diversity in human speech expression [10]. Therefore, other approaches need to be considered for robust spoken TOD models.

Speech Recognition Error Correction: Another area of research has concentrated on the interface between the speech recognition module and the DST module. This involves mapping spoken language to written language, an approach commonly referred to as speech error correction [22, 23]. The goal is to convert the spoken transcript into a written form that can be more easily processed by the existing text-based DST module. However, this method tends to strip away essential semantic information, as the transformation from spoken to written language can omit the subtleties and contextual cues present in natural speech. For example, the utterance “*I want to book a hotel for three people*” in written style is likely to be changed to an indirect dialog style in spoken scenarios like “*I want to book a place to stay for me and my parents*”. Correctly inferring the intent as *hotel-booking* and *slot as three-people* requires the underlying text model to be trained on a spoken reasoning style.

Text-Based Augmentation: To reduce dependence on speech-based data, researchers have explored text-based augmentation techniques, such as rule-based paraphrasing and slot augmentation from ontology [23, 24]. While these approaches introduce spoken-style variations in dialog utterances, they don’t add spoken-style indirect reasoning and the complex multi-turn information collection in TOD.

Our work utilizes LLMs to synthesize spoken-style TOD data while not only synthesizing the spoken-style semantic reasoning at the utterance level but also incorporating the multi-turn slot acquisition part of the dialog. In the following section, we will explain the core of TOD systems, their central component DST and its adaption in spoken scenarios.

4. Proposed Spoken TOD Data Synthesis

LLMs have demonstrated an extraordinary ability to generate synthetic data in various domains, including TOD [25]. However, in spoken scenarios, TOD doesn’t only need turn-based spoken style utterance but also includes aspects of indirect reasoning and multi-turn slot acquisition, as Figure 3 shows the multi-domain TOD example from a recently proposed spoken TOD benchmark SpokenWOZ [7]. Each dialog section has its

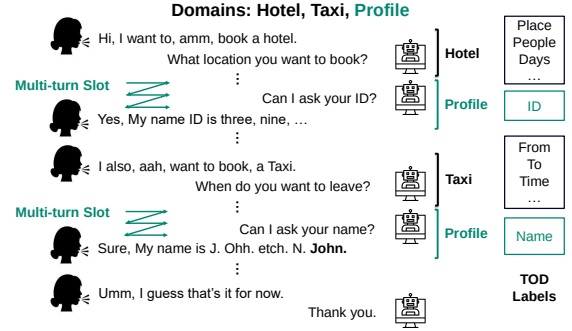


Figure 3: Dialog structure of the reference spoken-style TOD data SpokenWOZ [7]. The Profile domain requires a multi-turn dialog for a single slot collection.

corresponding TOD labels, including intent, slot, and values. The fully labeled TOD dialog file becomes excessively large and is not possible to be transformed in spoken style in an end-to-end way using LLM prompting. In this work, we employed a divide-and-conquer approach to break down the spoken TOD generation in the form of Prompt1 and Prompt2.

4.1. Query Style Transfer (Prompt1):

As the non-profile domains of the reference spoken-style data share the same structure as the written-style data like MultiWOZ, it is possible to borrow the TOD labels as it is, while style-transferring the dialog utterances to spoken-style. After close analysis of the human-labeled benchmark SpokenWOZ [7], we listed three main properties of spoken-style dialogs that need to be simulated in synthesized dialog i.e., semantic reasoning, the addition of disfluencies due to filler words, and simulating speech recognition errors. We observed that LLM outputs were inconsistent in simulating all scenarios in the synthesized data. Especially, LLM failed to add semantic reasoning scenarios consistently, and the speech recognition errors were possibly corrected by LLM post-processing filters before generating the final output. Having failed to fully style-transfer the whole dialog even after providing up to 4 examples in the few-shot prompt format, we tried the Chain of Thoughts (CoT) [26] approach, and it consistently simulated all three properties of spoken-style utterances.

Chain of Thoughts: We employed CoT prompting to add semantic reasoning, spoken disfluencies through filler words, and speech recognition errors incrementally. In spoken dialog utterances, semantic reasoning occurs in three ways e.g. indirect information, time-based reasoning, and mathematical reasoning. We picked two examples for each scenario and included them in the LLM prompt. To simulate speech disfluencies by adding filler words we gave two representative examples to LLM and relied on its capability to generate dialog with a similar style. Lastly, for speech misrecognition, we picked another two representative examples from reference transcripts with recognition errors and asked LLM to add at least two speech recognition errors in each utterance.

4.2. Query Generation (Prompt2)

The multi-turn slot value acquisition dialog data doesn’t exist in written dialogs, hence the same principle of query transfer doesn’t apply. Instead, a new query generation approach is re-

Table 1: Statistics of written-style TOD data MultiWOZ, Spoken style TOD data SpokenWOZ, and our LLM-generated TOD data GenWOZ. S: Semantics, N: ASR Noise, T: Multi-turn slots

Data	Domain Count	Turns	Cross Turns	Slots
MultiWOZ	7	115,424	0	56,628
SpokenWOZ	8	203,074	16,870	33,600
GenWOZ _(S)	8	115,424	0	56,628
GenWOZ _(S+N)	8	115,424	0	56,628
GenWOZ _(S+N+T)	8	162,614	35,746	75,504

quired. [27] introduces a query retrieval approach for counterargument generation for a given argument, with a few-shot example in LLM. As in the case of multi-turn profile slot acquisition, we have target slot values from the ontology of reference SpokenWOZ [7] data. We employed the query generation approach by giving a few examples of a multiturn dialog and asking LLM to generate the corresponding dialog.

$$query = (t_1, q_1), (t_2, q_2), (t_x, \dots)$$

$$q_x = LLM(context, query, instruction)$$

Existing written-style benchmark, MultiWOZ[28] has seven domains i.e. *Train, Taxi, Hotel, Restaurant, Attraction, Hospital, Police*, and the SpokenWOZ has an additional domain, *Profile*, that requires multiturn dialog for single slot collection as listed in Table 1. As this dialog also has fine-grained labels of intent, slots, spans, and turn IDs, we employed a tabular data generation [29] approach, where LLM generates the dialog query along with all corresponding TOD labels.

In this work, we used LLaMA3.1[30] for data generation, a 405B parameters model, the highest performing open-source LLM at the time of data generation. As illustrated in Figure 1, after various trails of end-to-end and a few-shot prompting, we succeeded in synthesizing data in two parts utilizing CoT prompting and query generation. We combine the synthesized data with the existing reference data SpokenWOZ [28] and refer to our data as GenWOZ. The full data generation took 86 hours with LLaMA3.1 running on 8x NVIDIA H100 GPUs.

Table 2: Evaluation of different TOD benchmarks for Joint Goal Accuracy (JGA) using UBAR and ChatGPT. S: Semantics, N: ASR Noise, T: Multi-turn slots from Profile domain.

Method		Training Dialogs	JGA	
			w cross-turn	w/o cross-turn
MultiWOZ		9438	-	56.0
SpokenWOZ Baseline		4200	23.96	29.13
MultiWOZ + SpokenWOZ		13638	24.29	30.07
Prompt1	GenWOZ _(S)	13638	24.42	31.77
Prompt1	GenWOZ _(S+N)	13638	25.03	32.32
Prompt1 + Prompt2	GenWOZ _(S+N+T)	13638	25.90	32.52
ChatGPT (GPT4o-mini)		-	10.92	13.47
ChatGPT (GPT4o)		-	11.14	13.94

4.3. Training of the Dialog State Tracking Model

The newly generated synthetic data, designed to mirror the nuances of natural spoken language, is utilized to train a text-based DST model, UBAR [31]. UBAR is a GPT2-based language model trained on the written-style TOD task. The UBAR is finetuned with the default setting of UBAR, using AdamW optimizer with an initial learning rate of 1^{-3} . Fine-tuning UBAR for 25 epochs takes around 8 NVIDIA A100 GPU hours.

5. Evaluation of the Model

Recently, the SpokenWOZ [7] benchmark for spoken task-oriented dialog systems was released, which is the spoken version of the famous text-based benchmark, MultiWOZ [28]. As this benchmark is the first of its type, there is a huge performance gap between spoken dialog systems' accuracy compared to the written style dialog benchmark.

5.1. Experimental Results

Table 2 lists the results of JGA for our experiments on the SpokenWOZ [7] baseline. As explained in the earlier sections, we divided the data generation pipeline into two parts by utilizing Prompt1 and Prompt2. Evaluation of Prompt1 is further divided into two parts i.e. semantics of the dialog which includes reasoning and spoken style and the speech recognition noise simulation. Our experiments result on the UBAR model show that adding semantics alone in the transformed spoken style data achieved nearly 2.64% improvement in JGA in the case of without predicting the cross-turn label. Our experiments show that adding speech recognition errors in generated data improved the downstream DST model performance by 0.55%.

We then performed experiments by combining the data generated with Prompt1 and Prompt2. As the experimental results show, it further improved the JGA performance by 0.87% in the case when TOD model needs to predict the cross-turn label. We also compared the performance of the task-specific trained TOD model with the popular LLM framework ChatGPT using the TOD template-filling prompting. It is worth noting that ChatGPT struggled to understand a complex task like DST and delivered much lower performance on JGA accuracy. This shows that small and specialized models trained for a specific task perform better than a large general-purpose LLM. Overall, our experiments show that our proposed approach leads to an overall 3.39% absolute performance improvement and 11.6% relative improvements in JGA compared to the baseline.

6. Conclusion

Our work addresses the challenge of adapting text-based TOD systems for spoken scenarios. We have demonstrated that LLMs can improve the accuracy of spoken dialog systems by generating task-specific data. We also confirmed that training the model on noisy data that simulates the noise from the actual scenarios leads to improved performance. We also demonstrated how we can break the data generation tasks into smaller tasks for long dialog data generation like TOD. Overall, by generating spoken data using LLMs, the TOD model achieved a 3.39% absolute and 11.6% relative performance improvements in JGA. In the future, we plan to further increase the generated data size to develop more robust spoken TOD systems.

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